The Central Distribution: A Noninformative Prior for Feature Selection

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1 Introduction

1.1 The Optimal Strategy

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Furthermore, we assume that an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_{S} \sum_{a} \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e),$$

where P^* is the joint probability distribution of the environment state and the evidence.

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

A strategy that, for each evidence e, selects the action minimizing the expected loss conditioned on e, is optimal:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\min_a \mathbb{E}_{P^*} \left[\ell(a, x) \mid e\right] \\ &= \arg\min_a \sum_x \ell(a, x) P^*(x \mid e) \cdot \end{aligned}$$

Now, suppose that we decide to use a different evidence E' instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that, in reality, we cannot freely choose any random variable as our evidence. We can only choose evidence that is a deterministic function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E'. Thus, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory. However, in practice we rather to use feature selection because without it the original learning task requires impractical large datasets.

The key point here is that the space of strategies using E' is a subset of the strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well.

We believe that this is why the Bayesian framework does not favor the simpler evidence E'. Plausibly, if we choose an appropriate prior distribution, when we are finding the optimal strategy, feature selection should be performed automatically within the Bayesian framework. However, many commonly used non-informative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for finding a noninformative prior that also exhibit feature selection characteristics.

1.2 Divergence

Suppose that for a given environment, we have specified a probability measure P. In particular, we are interested in the distribution it induces on a random variable X, denoted $P(X)$.

Now, imagine we ask an expert—a person familiar with the environment—to evaluate our choice of P. The expert may hold different beliefs and instead favor another probability measure Q. From the expert's perspective, evaluating P involves assessing the potential loss incurred by using the (in their view) incorrect distribution $P(X)$ when the true distribution is believed to be $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose that $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$ is some divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable, and Q be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left[Q(\Omega) \,\|\, U(\Omega)\right] = \mathbb{H}\left[U(\Omega)\right] - \mathbb{H}\left[Q(\Omega)\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y and probability measures P, Q*

$$\mathbb{D}\left[Q(X,Y) \parallel P(X,Y)\right] = \mathbb{D}\left[Q(X) \parallel P(X)\right] + \sum_x \mathbb{D}\left[Q(Y|x) \parallel P(Y|x)\right] Q(x) \cdot$$

If $\mathbb{D}[\,\cdot\,]$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise} \cdot \end{cases}$$

Notice that both $P(\Omega|X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P(\Omega|x)\right] = \ln P(x)$, $\mathbb{H}\left[P(\Omega, X)\right] = 0$.

We define $Q(\Omega|x)$ to be an arbitrary distribution that has the same support as $P(\Omega|x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] &= \mathbb{H}\left[P(\Omega, X)\right] - \mathbb{H}\left[Q(\Omega, X)\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned} \mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] &= \mathbb{H}\left[P(\Omega|x)\right] - \mathbb{H}\left[Q(\Omega|x)\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega \cdot \end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned} \mathbb{D}\left[Q(X) \parallel P(X)\right] &= \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] - \sum_x \mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)} \cdot \end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(XY)$ from $Q(XY)$ is a measure of the generic loss of using $P(XY)$ when we believe $Q(XY)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

We now briefly review some KL divergence properties needed in subsequent sections.

Lemma 1. *Let X be a random variable, and let P, Q, and R be probability distributions over X. The Kullback–Leibler divergence satisfies the following inequality:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \geq \sum_x Q(x) \ln \frac{R(x)}{P(x)} \cdot$$

Proof. R is a probability distribution, and the Kullback-Leibler divergence is non-negative:

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, R(X)\right] \geq 0 \cdot$$

By adding $\sum_x Q(x) \ln \frac{R(x)}{P(x)}$ to both sides of the above inequality, the desired inequality follows. □

Theorem 2. *Let Q_1, Q_2 and P be probability distributions. If for some $0 \leq \alpha \leq 1$ we have $Q = \alpha Q_1 + (1-\alpha) Q_2$, then:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \leq \alpha \mathbb{D}_{\mathrm{KL}}\left[Q_1(X) \,\|\, P(X)\right] + (1-\alpha) \mathbb{D}_{\mathrm{KL}}\left[Q_2(X) \,\|\, P(X)\right] \cdot$$

That is, the KL divergence is convex in its first argument.

Proof. The result follows from Lemma 1 after expanding the left-hand side of the inequality using the definition of KL divergence. □

The KL divergence plays a key role in the notion of central distribution, which we introduce in the next section.

2 Central Distribution

Consider a random variable X. In probabilistic modeling, our knowledge about X is typically represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may only be able to specify a set of plausible distributions. This set can be viewed as a representation of *partial knowledge*, or equivalently, a state of lack of knowledge or *ignorance*.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \ldots$ is said to be exchangeable, if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively narrowing the set of admissible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the perspective that such sets provide a meaningful representation of the state of partial knowledge, and that this representation can likewise be used for inference.

Importantly, the act of choosing this set reflects some degree of knowledge. If even that is uncertain, we may similarly represent our uncertainty by a set of sets of distributions—capturing deeper layers of ignorance.

In Bayesian theory, any type of knowledge must ultimately be encoded in a single probability measure. Thus, to use our partial knowledge for performing inference in a Bayesian manner, we must find a principled way to convert our set-valued representation into a single probability measure.

The probability measure that we choose, represents our beliefs about the environment. It can be argued that this choice must depend solely on the environment. As long as the environment is the same, we must not change this choice. For example, if our loss function changes, or if the evidence that we get from the environment becomes more or less informative, none of these should change our beliefs about the environment, and therefore our chosen probability measure. We will use this principle several times in this discussion and refer to it as the *principle of objectivity*.

Subjectively, when we are uncertain about a set of distributions, a probability measure that has the lowest divergence from the distributions of the set can be considered the "safest" choice. "Safe" in the sense that using it incurs the least possible loss in the worst case, and more importantly, when convergence is possible for a set of distributions, if we use this probability measure, we are guaranteed to have a convergent distribution. By convergent distribution, we mean a distribution with sufficiently low divergence from any other distribution in that set.

Definition 2.1 (Central Distribution)**.** Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X. We say a probability distribution C is a *central distribution* of $\mathcal{Q}$ for X, if

$$C \in \arg\min_{P} \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel P(X) \right], \tag{1}$$

where $\mathbb{D}_{\mathrm{KL}}$ denotes the Kullback–Leibler (KL) divergence.

Note that, under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Definition 2.2 (Interior Distribution)**.** Let $\mathcal{Q}$ be a set of probability distributions over a random variable X. A distribution R is called an *interior distribution* of $\mathcal{Q}$ if, for every probability distribution P, there exists some $Q \in \mathcal{Q}$ such that

$$\mathbb{D}_{\mathrm{KL}} \left[R(X) \parallel P(X) \right] \leq \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel P(X) \right],$$

Theorem 3. *Any distribution that can be expressed as a convex combination of some elements of $\mathcal{Q}$ is an interior distribution of $\mathcal{Q}$.*

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$. Fix an arbitrary probability distribution P. By the convexity of KL divergence in its first argument, we have:

$$\mathbb{D}_{\mathrm{KL}} \left[R(X) \parallel P(X) \right] \leq \sum_i \alpha_i \mathbb{D}_{\mathrm{KL}} \left[Q_i(X) \parallel P(X) \right] \cdot$$

Now, note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index j such that

$$\mathbb{D}_{\mathrm{KL}} \left[R(X) \parallel P(X) \right] \leq \mathbb{D}_{\mathrm{KL}} \left[Q_j(X) \parallel P(X) \right] \cdot$$

This satisfies the requirement of the definition, and R is indeed an interior distribution of $\mathcal{Q}$. □

Theorem 4. *Let R be an interior distribution of $\mathcal{Q}$. If C is a central distribution of $\mathcal{Q}$, it is also a central distribution of $\mathcal{Q} \cup \{R\}$ and vice versa.*

Proof. This theorem follows directly from the definition of an interior distribution, which implies that for any probability distribution P, we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel P(X) \right] = \sup_{Q \in \mathcal{Q} \cup \{R\}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel P(X) \right] \cdot$$ □

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

Theorem 5. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T). Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T)Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T, then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T).

Proof. Let C be a central distribution of $\mathcal{Q}$ for T. For any $P \in \mathcal{Q}$ we have

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, R(X \mid T)C(T)\right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, R(X \mid T)P(T)\right] \cdot \quad (2)$$

Since the KL divergence is nonnegative, for every probability distribution P and every $Q \in \mathcal{Q}$, we have

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, R(X \mid T)P(T)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, P(X,T)\right] \cdot$$

That implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, R(X \mid T)P(T)\right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, P(X,T)\right] \cdot$$

Therefore, combining this inequality with Inequality 2, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T). □

Intuitively, a central distribution serves as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall behavior of the set in a way that is meaningful for inference or comparison with other distributions.

Definition 2.3 (Central Divergence)**.** For any distribution P, we define the *central divergence* of P from a set of distributions $\mathcal{Q}$, with respect to a random variable X, as follows:

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q} \,\|\, P(X)\right] = \inf_{C} \mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, P(X)\right],$$

where C denotes a central distribution of $\mathcal{Q}$ for X.

Note that when $\mathcal{Q}$ has a unique central distribution, the central divergence is simply defined as the KL divergence from that unique central distribution.

Definition 2.4. Let X be a random variable, and let $\mathcal{Q}^*$ be a set whose elements are sets of distributions over X. Let $C_{\mathcal{Q}}$ be a central distribution of $\mathcal{Q} \in \mathcal{Q}^*$ for X, and P be an arbitrary probability distribution. We say C^* is a central distribution of $\mathcal{Q}^*$, if we have:

$$\sup_{\mathcal{Q} \in \mathcal{Q}^*} \mathbb{D}_{\mathrm{C}} \left[\mathcal{Q} \, \| \, C^*(X) \right] \leq \sup_{\mathcal{Q} \in \mathcal{Q}^*} \mathbb{D}_{\mathrm{C}} \left[\mathcal{Q} \, \| \, P(X) \right] \cdot$$

Definition 2.5 (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f, if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Theorem 6. *Let X be a discrete random variable, and $\mathcal{Q}$ be a finite set of distributions for X. Consider the following function:*

$$f(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where $Q_i \in \mathcal{Q}$, and P, R are probability distributions.

If (P^*, R^*) is a saddle point of f, then P^* is a central distribution of $\mathcal{Q}$ for X.

Proof. Since X is a discrete random variable, $\mathcal{Q}$ has a central distribution for X. Let C be a central distribution of $\mathcal{Q}$, and (P^*, R^*) be a saddle point for f. Clearly, the point (C, R^*), is in the domain of f, and since (P^*, R^*) is a saddle point for f, we have

$$f(P^*, R^*) \leq f(C, R^*) \cdot \tag{3}$$

Because R^* is a probability distribution, the definition of f implies:

$$f(C, R^*) \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \, \| \, C(X) \right] \cdot$$

On the other hand,

$$f(P^*, R^*) = \max_R f(P^*, R) = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \, \| \, P^*(X) \right] \cdot$$

Thus, Inequality 3 implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \, \| \, P^*(X) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \, \| \, C(X) \right] \cdot$$

Consequently, P^* is also a central distribution of $\mathcal{Q}$ for X. $\square$

Lemma 2. *Let f be the function defined in theorem 6. The point (P^*, R^*) is a saddle point for f, if and only if there are constants λ, η and a function μ, such that for every i, the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \tag{4}$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \tag{5}$$

$$\sum_i R^*(i) = 1, \tag{6}$$

$$R^*(i) \geq 0, \tag{7}$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0 \cdot \tag{8}$$

Proof. Similar to f, we define $\hat{f}$ as follows:

$$\hat{f}(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where P, R are arbitrary real functions instead of probability distributions. This enables us to formulate the saddle point of f as the solution to a constrained optimization problem.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & \hat{f}(P, R) & \text{w.r.t. } P \\ \text{subject to} & \sum_x P(x) = 1, & \\ & & \\ \text{maximize} & \hat{f}(P, R) & \text{w.r.t. } R \\ \text{subject to} & R(i) \geq 0, & \text{for all } i \\ & \sum_i R(i) = 1 & \end{array} \tag{9}$$

Clearly, if (P^*, R^*) exists, it will be a saddle point for f.

Both of these optimization problems are convex [4] and differentiable. To establish this, consider that all the constraint functions are affine, and therefore are convex and differentiable.

Additionally, $\hat{f}$ is differentiable function with respect to both R and P. For any fixed P, $\hat{f}(P, \cdot)$ is an affine function and is concave.

On the other hand, for a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P, \hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x), \tag{10}$$

where c is not a function of P. Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the constraints of the second problem and therefore is a probability distribution. Hence, for all i, x we have $\hat{R}(i)Q_i(x) \geq 0$, and $\hat{f}(P, \cdot\,)$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [4]. In other words, If a point (P^*, R^*) satisfies the KKT conditions for both problems, it will be a saddle point for f, and vice versa.

The KKT conditions are as follows for every x, i and constants λ, η:

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \tag{11}$$

$$\sum_x P^*(x) = 1, \tag{12}$$

$$\begin{gathered} \sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \\ \sum_i R^*(i) = 1, \\ R^*(i) \geq 0, \\ \mu(i) \geq 0, \\ \mu(i) R^*(i) = 0\cdot \end{gathered} \tag{13}$$

By summing Equation 11 over x and using Equation 13, we have

$$\eta = 1\cdot$$

Thus, we get Equation 5 which also implicitly implies Equation 12. $\square$

Theorem 7 (Centrality)**.** *Let X be a discrete random variable, and let $\mathcal{Q}$ be a finite set of distributions over X. Suppose C is a convex combination* *of elements of $\mathcal{Q}$, with all weights strictly positive. If for every $Q \in \mathcal{Q}$ and some constant λ we have:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] = \lambda,$$

then C is a central distribution of $\mathcal{Q}$ for X.

Proof. Lemma 2 provides a set of sufficient conditions for a saddle point. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions. We can replace Inequality 7 with the more restrictive $R^*(i) > 0$ condition. By Equation 8, that implies $\mu(i) = 0$, and Lemma 2 conditions can be simplified to

$$\sum_x Q(x) \ln \frac{Q(x)}{P^*(x)} = \lambda,$$
$$P^*(x) = \sum_i R^*(i) Q(x),$$

which must hold for every $Q \in \mathcal{Q}$ and a positive probability distribution R^*.

Since C satisfies these condition, Lemma 2 and Theorem 6 imply that C is a central distribution of $\mathcal{Q}$ for X. □

These theorems were proved under the assumption that $\mathcal{Q}$ is finite; however, they can be extended to cases where $\mathcal{Q}$ can be indexed by a finite-dimensional real vector Θ. We can treat this vector as a random variable and interpret Q_θ as a conditional distribution $Q(x \mid \theta)$. Then, the weight function, $R^*(i)$, will essentially become a prior distribution for θ.

Theorem 8. *Let X be a discrete random variable and $\mathcal{Q}$ be a set of distributions over X, indexed by a finite-dimensional random vector Θ. Let c be a positive prior distribution for Θ such that for $Q(x \mid \theta) \in \mathcal{Q}$ we have*

$$c(\theta) \propto \exp\left\{ \sum_x Q(x \mid \theta) \ln c(\theta \mid x) \right\}.$$

Then, $C(x) = \int Q(x \mid \theta) c(\theta) d\theta$ is a central distribution of $\mathcal{Q}$ for X. We also refer to c as a central prior for X.

Proof. To Do!!! □

Suppose we wish to model an exchangeable sequence of random variables. Let S_n denote the sequence of the first n variables: $S_n = (X_1, X_2, \ldots, X_n)$.

As discussed earlier, exchangeability can be interpreted as a form of partial knowledge, which allows us to represent our uncertainty using a central distribution over S_n.

However, this central distribution generally depends on the sequence length n. According to the principle of objectivity, our beliefs should not depend on the specifics of our experimental setup, and in particular, should not vary with the length of the sequence.

It is important to note that a distribution defined over a longer sequence induces marginal distributions over its shorter subsequences. This observation suggests that, in order to satisfy the principle of objectivity, we may instead determine a central distribution over an infinite exchangeable sequence, from which the distributions of all finite initial segments can be derived.

This requires constructing a central distribution that is well-defined over S_∞.

Definition 2.6. Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Assume C_n is a central distribution for S_n. Let C be a probability measure that defines a distribution over every S_n. C is a central probability measure of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$, if we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0\cdot$$

Intuitively, this definition states that as $n \to \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n. In other words, C is a central distribution for S_∞.

If we have a consistent estimator sequence for the index of a set, the following theorem provides a way to construct a central distribution for that estimator.

Here, by a consistent estimator we mean a sequence of estimators such that, for any fixed index value, the distribution of each estimator T_n becomes increasingly concentrated around the corresponding value of the index as n increases.

Theorem 9. *Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of random variables, and let $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each T_n. Assume that $\mathcal{Q}$ can be indexed by a finite-dimensional vector Θ. If $\{T_n\}_{n=1}^{\infty}$ is a consistent sequence of estimators for Θ, and there exists a sequence of constants $\{\lambda_n\}_{n=1}^{\infty}$ such that the sequence $\{\lambda_n - \mathbb{H}\left[Q(T_n \mid \theta)\right]\}_{n=1}^{\infty}$ converges for every value of θ, then*

$$c(\theta) \propto \lim_{n\to\infty} \exp\left\{\lambda_n - \mathbb{H}\left[Q(T_n \mid \theta)\right]\right\},$$

is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$.

Proof. To Do!!! □

A central distribution over an estimator cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorem 5 provides a simple way for carrying out this conversion.

As an illustrative example, consider a binomial experiment. Theorem 9 allows us to determine a central distribution over the number of successes. However, for inference, we typically require a distribution over specific ordered sequences of successes and failures. Theorem 5 enables this by offering a natural approach: we assign a uniform distribution over all sequences with the same number of successes and failures.

When dealing with layered ignorance and attempting to determine the central distribution of a set of sets of distributions, as defined in Definition 2.4, we typically consider the case where the number of such sets is finite.

In this setting, the central distribution is simply the uniform distribution.

Corollary 9.1. *Under the assumptions of Theorem 9, if $\mathcal{Q}$ is a countable set indexed by a discrete random variable I, then $c(i) \propto \lambda$ is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$, where λ is a constant.*

Example 2.1 (Fused Multinomial Model)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence where each $S_n = (X_1, X_2, \ldots, X_n)$ represents n independent trials of an experiment with k possible outcomes. Let $f : \mathcal{X} \to \mathcal{Y}$ be a deterministic function that maps each outcome X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \ldots, m\}$.

Assume that given $Y_i = y$, the conditional distribution $P(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$. Let $\theta = (\theta_1, \ldots, \theta_m)$ be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y.

For a sequence S_n, let n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f. Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$ is:

$$Q(S_n \mid \theta) = \prod_{y=1}^{m} \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Now, consider the estimator sequence $T_n = \frac{1}{n}(n_1, n_2, \ldots, n_m)$, which is a consistent estimator for θ. The conditional distribution $P(S_n \mid T_n)$ is then given by:

$$P(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^{m} \frac{n_y!}{|f^{-1}(y)|^{n_y}}.$$

Note that this distribution is independent of the parameter θ. Therefore, by Theorem 5, a central distribution for S_n can be obtained from a central distribution for T_n.

As $n \to \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Theorem 9, we can determine a central prior for $\{T_n\}_{n=1}^{\infty}$:

$$c(\theta) \propto \prod_{y=1}^{m} \theta_y^{-\frac{1}{2}}$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n:

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right)\Gamma\left(n+1\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma\left(y+1\right)}.$$

Note that here, C denotes a central distribution for T_∞.

By Theorem 5, we can derive a central distribution for $\{S_n\}_{n=1}^{\infty}$, which induces the following distribution over each S_n:

$$C_f(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)|f^{-1}(y)|^{n_y}}.$$

Example 2.2 (Attribute Selection)**.** In this example, we consider a classification problem where the goal is to predict a target class, denoted by Y, from a set of observed nominal attributes $(A_1, A_2, \ldots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

This reflects a form of sparsity in the relationship between the class and the attributes, where irrelevant features are ignored in the true generative model.

We can use the fused multinomial model of Example 2.1 to capture this relationship. As an example, assume that Y depends only on A_2, A_5. Then, we define the labeling function f as follows:

$$f(A_1, A_2, \ldots, A_n, Y) = (A_2, A_5, Y)\cdot$$

In a fused multinomial model with this labeling function, Y will be conditionally independent from other attributes, given A_2, A_5. Additionally, $P(A_1, A_2, \ldots, A_n \mid A_2, A_5)$ will be uniform.

For every possible subset of attributes a distinct fused multinomial model of this form is needed. Each model corresponds to a set of plausible distributions. Definition 2.4 provides a way to define a central distribution for situations like this one.

By Theorem 7 we know that this central distribution would be a weighted average of central distributions of individual fused multinomial models. Since, we have a finite number of models, Corollary 9.1 implies that these weights are uniform:

$$C_{\mathcal{Q}^*}(S_n) \propto \sum_i C_{f_i}(S_n).$$

To establish this, observe that we can construct a consistent estimator for the model indicator I by checking for the equality of different outcomes. Specifically, the estimator identifies outcomes with equal frequency in the data and returns the indicator for the model that supports this equality.

To see how this idea works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each feature independently with equal probability for 0 and 1. The class label was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same relevant features and labeling function were used across all trials.

For each experiment, we generated a training set of size 20 and a test set of size 60. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant features.

At each step, we trained and tested two versions of the central distribution: one assuming a deterministic label function (CD-Det) and one allowing for a conditional (non-deterministic) label distribution (CD-Prob). As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 75 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, shown in Figures 1 and 2, reveal notable differences in performance depending on the underlying class assignment rule. For function f_1, both deterministic and probabilistic methods achieve high accuracy, with the deterministic version of the central distribution slightly outperforming the other. In contrast, function f_2 introduces more ambiguity in the label assignment, leading to a clearer separation in performance between these two methods. In this case, the deterministic central distribution remains the most

Table 1: Mappings of 3-bit inputs under functions f_1 and f_2

Input (bits)	f_1(Input)	f_2(Input)
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

Figure 1: Accuracy vs. Number of Attributes (Function f_1, 20 train samples)

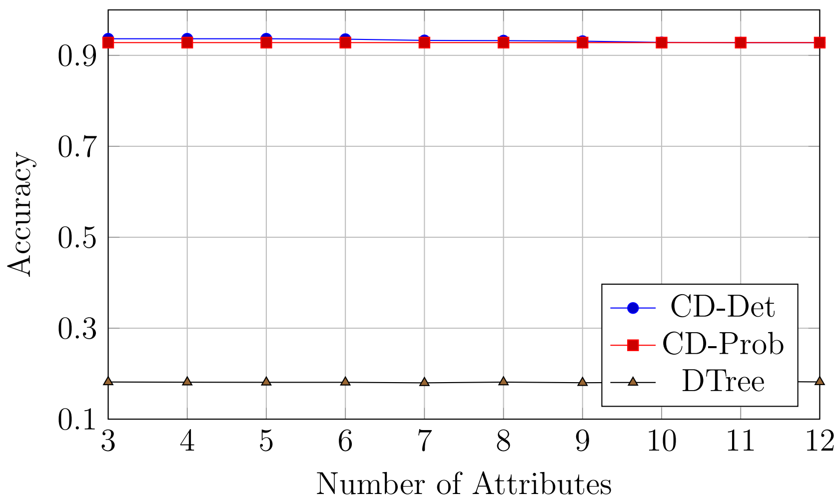
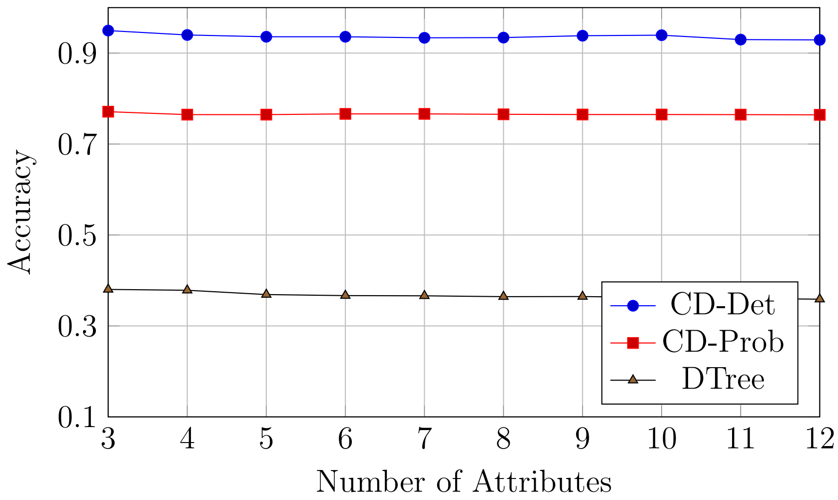
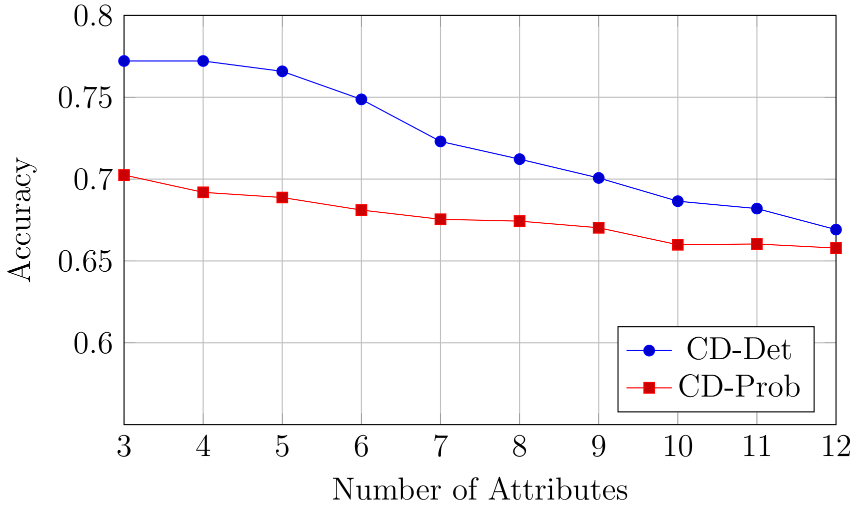


Figure 2: Accuracy vs. Number of Attributes (Function f_2, 20 train samples)



accurate overall, while the probabilistic version shows reduced performance due to label overlap.

Figure 3: Accuracy vs. Number of Attributes (Function f_2, 10 train samples)



While removing irrelevant attributes generally led to slight improvements in accuracy, the effect was not substantial in our primary experiments with 20 training samples. However, we observed that the benefit of attribute removal becomes more noticeable when the training set is smaller. In particular, additional experiments with only 10 training samples showed a clearer performance gain as irrelevant features were removed, indicating that the impact of irrelevant attributes is more pronounced in low-data regimes (see Figure 3)

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